Overview

For your final project, you will design and implement a robotics solution. *This is your opportunity to experiment, think creatively, and demonstrate everything you've learned throughout the course.*

You have two project options:

1. **Default Project** – Imitation learning for the Open Cabinet task.
2. **Open Project** – Propose and pursue your own original idea.

You may have 2 team members per team for the default project and a maximum of 4 for the open project.

Grading (% out of total course grade):

- Project Report: 8%
- Website with Demo Video: 5%
- Source Code: 2%
- In-class project briefing (extra credit) : 1%

Deliverables:

Final project report (8%)

- Problem is clearly defined 1.5%
- Method
  - is clearly described 1%
  - is technically sound 2%
- Evaluation results
  - is well presented 1%
  - met success criteria 1%
- Included discussion and reflections 1%
- Discusses team contribution 0.5%

Project Website with Demo Video (5%)

- A clear, visually engaging summary of your project (2%)
  May include images, interactive elements, or additional context
- A video clearly demonstrate your solution in action (3%)

Source Code 2%

- Well-documented and organized
- Includes a comprehensive **README**
- Fully **reproducible**

GenAI Usage for Final Project *

- You are allowed to use any tool of your choice and build on existing repositories as long as you properly credit the source.
- However, you need to define your contribution clearly (i.e. what is the code/functionality you implement).

Option 1: Default Project — Robocasa Open Cabinet Door Task

Objective: Implement an imitation algorithm to solve the **Open Cabinet Door** Task using provided demonstrations.

Starter codebase: https://github.com/HoldenGs/cs188-cabinet-door-project

Project Expectations:

- Train a policy using supervised learning on provided demonstration data.
- Discuss challenges, limitations, and potential improvements in the report

Option 2: Open Project - Your idea!

We encourage you to pursue an open project if you're interested in building something creative, ambitious, or personally meaningful. This is your chance to take everything you've learned and apply it to an idea you're passionate about.

Your project must include a robotics component, but beyond that, you're free to explore any tools, platforms, or topics that inspire you. Whether it's artistic, interactive, research-driven, or just something fun—go for it!

Just make sure to get your idea approved by the instructor or a TA.

Example topics:

- **AI Agent for Robotic Control:** Implement an AI agent that converts high-level natural language instructions into executable robot actions in robosuite. The system must: (1) parse at least 3 distinct task instructions (e.g., "stack the blocks," "sort by size"), (2) generate a sequence of low-level actions or subgoals, and (3) successfully execute the task in simulation. Include a clear mapping from language → plan → control.
- **Shared Control (Human + Autonomy):** Design a shared-control system in robosuite where the user controls at least one degree of freedom (e.g., end-effector position via joystick), while an autonomous policy controls another component (e.g., grip force, vertical stabilization, or obstacle avoidance). Demonstrate improved task performance or usability compared to manual-only control.
- **Learning to Act (Reinforcement Learning):** Train a robosuite robot using a standard RL algorithm (e.g., PPO, SAC) to complete a defined manipulation task (e.g., door opening, nut assembly, pick-and-place). The project must include: (1) task definition, (2) reward design, (3) training curve visualization, and (4) quantitative evaluation of success rate.
- **Multimodal Control (Gesture or Voice):** Implement a gesture-based or speech-based interface that triggers at least 3 distinct robot actions in robosuite (e.g., "pick up," "stop," "reset"). The system must include input processing

(gesture recognition or speech-to-text) and a mapping from detected commands to robot control policies. Provide a live or recorded demo.

- **Expressive Motion (Emotional Trajectories):** Design and implement at least 3 distinct expressive motion styles for a robot arm (e.g., excited, sad, neutral) by modifying trajectory speed, smoothness, or acceleration profiles. Conduct a small user study (≥5 participants) to evaluate whether observers can correctly identify the intended emotion.
- **Social Cues (Simulated Attention):** Implement a gaze or orientation behavior in robosuite where the robot dynamically directs its end-effector or camera toward relevant objects, goals, or agents. Compare task clarity or perceived intentionality with and without gaze behavior, using short user feedback or qualitative analysis.
- **Dancing Robot (Music Synchronization):** Implement beat detection for an audio track and synchronize robot joint or end-effector motion to the detected tempo. The system must: (1) extract tempo automatically, (2) generate time-aligned motion patterns, and (3) demonstrate synchronization accuracy in simulation.

Or ANYTHING related to a robot that you think is cool to work on!!!! :)